Detached Bridgman Growth—A Standard Crystal Growth Method with a New Twist

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Abstract

Bridgman or vertical gradient freeze (VGF) crystal growth processes have several advantages compared to other melt growth methods, especially the possibility to achieve a low level of thermal stress and low dislocation densities in the grown crystals. However, crystals grown in contact with a crucible usually suffer from mechanical stress during cooling, reducing the structural quality. The "detached" or "dewetted" Bridgman growth avoids this problem and has recently been investigated in more detail as a promising tool to improve crystal quality. Detached growth, where the crystal is separated from the crucible wall by a gap of 10–100 μm, was found originally in some microgravity experiments going back to 1975. Considerable improvements of crystal quality were reported for those cases; however, the reasons for the detachment were not fully understood. In the last 10–15 years, theoretical investigations as well as new experiments have shown beyond a doubt that detached growth can, in principle, be achieved in Earth's gravity with the same advantages that were demonstrated in the crystals grown under microgravity. It could be shown that the ability to achieve detachment depends on a complex interplay of the wetting of the melt with the crucible and the crystal as well as the pressure balance in the system, including the hydrostatic pressure, the gas pressure above the melt, and the pressure below the melt. It turns out that for stable detachment, only, specific combinations of meniscus shape, gap size, wetting angle, growth angle, and pressures work. The conditions that lead to detachment are thus highly specific for a given system.

Introduction

The three major melt growth methods of industrial importance for semiconductor crystal growth are the Czochralski (CZ) method, the float-zone (FZ) method, and vertical Bridgman growth, including related methods of directional solidification in a crucible such as vertical gradient freeze (VGF).

The CZ method entails dipping a seed crystal into a melt and pulling upward, usually while the seed crystal and/or the melt is rotated. A liquid encapsulant is sometimes used in this method to prevent the evaporation of lower melting point constituents of the material being grown. In the FZ method, a liquid zone moves through the material. The liquid zone is held in place by surface tension, not by an external container. The surface tension of typical melts limits the zone height and thus the maximum diameter of crystals grown by FZ to about 20 mm.

In the Bridgman process, the melt is contained inside of a crucible. The crucible is translated with respect to a temperature gradient. The crystal solidifies as the freezing point of the temperature gradient travels along the crucible. In the VGF method, the crucible is stationary, and the temperature field is shifted electronically; otherwise, it is similar to the Bridgman method. Bridgman and VGF growth are used on an industrial scale for the growth of III–V and II–VI semiconductors, multicrystalline silicon for solar cells, and calcium fluoride. The advantages compared to the CZ method are a relatively simple setup, with no mechanical movement during growth in the case of VGF, suitability for systems with comparatively high vapor pressures, the possibility to grow crystals with very large dimensions, a low level of convective flow in the melt, and especially a very good control of the temperature field with small radial gradients. This allows a significant reduction of thermal stress compared to CZ-grown crystals of the same size, resulting in crystals with low dislocation densities. For instance, in the case of III–V compounds, the reduction in dislocation density can be more than an order of magnitude for VGF-grown crystals compared to liquid encapsulation CZ (LEC)–grown crystals of the same type and diameter. One of the main disadvantages, apart from the impossibility of visually controlling the process, is the interaction of the crystal and the melt with the crucible material. Even if there is no direct chemical interaction of the melt with the crucible, the mechanical contact of the crystal with the crucible wall can lead to the formation of twins and heterogeneous nucleation during growth and the development of mechanical stress during cooling because of the different expansion coefficients of the crystal and the crucible material. This often negates the advantage of the lower temperature gradients during growth. While this problem can be partially reduced for a few selected material systems by liquid encapsulation (e.g., by the use of B_2O_3 for VGF growth of GaAs), no contact at all between the crystal and the crucible, as in CZ and FZ growth, would be preferable. This is possible in a configuration usually called "detached" or "dewetted" Bridgman growth.

In detached growth, most of the melt has contact with the crucible wall, similar to the regular Bridgman setup, but the crystal is separated from the wall by a small gap, typically of the order of 10^{-5} m. Detached growth was first found unexpectedly in microgravity crystal growth experiments of InSb and Ge on Apollo-Soyuz (1975) and Skylab (in orbit 1973–1979), and a significant reduction in crystal defects was reported for those crystals.[1,2] However, microgravity did not always lead to detachment in Bridgman crystal growth. The reasons for detachment, apart from the general

understanding that the absence of hydrostatic pressure played a role, were not really understood for quite a while, and the effect was considered more of a curiosity. A renewed interest in detachment, driven by the need to further improve the structural quality of crystals, started about 10–15 years ago. An overview of earlier work on detachment can be found in Wilcox and Regel.[3]

A schematic drawing of detached growth is shown in Figure 1. The crystal is removed from the wall and connected to the melt by a small meniscus. The two angles shown, θ and α, are the wetting angle of the melt with the crucible and the growth angle, respectively. It has been shown[4,5] that if there is no additional pressure difference in the system, including the hydrostatic pressure due to gravity, detachment will occur if

$$\theta + \alpha \geq 180° \quad (1)$$

(i.e., detachment is determined by the sum of the wetting angle and the growth angle). Because the growth angle α mostly depends on the material and the crystal orientation, it is essentially fixed for a given material; in the case of germanium and silicon, it is about 11°. This leaves only the wetting angle θ (i.e., the choice of the crucible material) as a degree of freedom for the crystal grower. The angle θ should obviously be as high as possible; because the growth angles of most semiconductors are in the range of 10–30°, wetting angles would need to be of the order of 150–170° to achieve dewetting based on this mechanism.

The introduction of pressures complicates the occurrence of detachment. First, one has to consider the hydrostatic pressure in the melt due to gravity and, second, a possible pressure difference between the volume below the melt (i.e., the gap) and the volume above the melt. Whereas the hydrostatic pressure is detrimental to detachment, the application of an overpressure in the gap compared to the pressure in the upper volume can promote detachment.

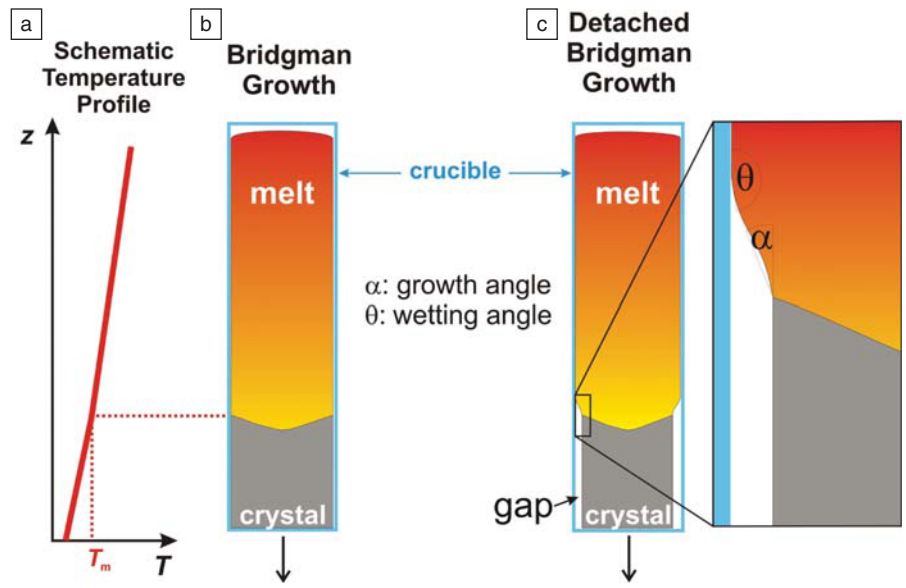

Figure 1. (a) Temperature profile together with (b) a schematic drawing of the regular Bridgman growth setup and (c) the detached Bridgman growth configuration. The label *z* merely indicates the vertical direction.

Theoretical Background

There have been several studies of the mechanisms that may lead to detached solidification.[6–11] Significant issues are the conditions required to establish a meniscus between the solid and ampoule at the solid/melt interface, the resultant menisci shapes, and the conditions under which established gap widths will remain stable. The essential features of the detached growth process are shown in Figure 2. Depicted is a cross section of a cylindrical ampoule with the central axis of the ampoule on the left-hand-side of the figure and the ampoule wall at a distance R_0 on the right. The melt is in contact with the ampoule wall, but the solid is not. As growth proceeds, the position of the solid and the meniscus move upward with respect to the ampoule. There is a pressure in the gas above the melt, P_H, and a pressure in the gas below the melt, P_C. The pressure difference across the meniscus is given by

$$\Delta P = \Delta P_0 - \rho g z = P_C - P_H + \rho g z - \rho g z, \quad (2)$$

where ΔP_0 is the pressure differential at the three-phase junction, ρ is the melt density, *g* is the gravitational acceleration, *h* is the melt height, and *z* represents the vertical coordinate measured from the top of the solid. The meniscus shape is determined by several factors: the pressure differential, the growth angle α, the wetting angle θ, also called the contact angle, and the curvature of radii r_1 and r_2. Here r_1 is in the plane of the figure, and r_2 is in a perpendicular plane with its origin on the sample axis. The Young-Laplace equation defines the relation between the pressure differential across the meniscus and the meniscus radii[7,10]

$$\Delta P = \sigma\left(\frac{1}{r_1} + \frac{1}{r_2}\right), \quad (3)$$

where σ is the surface tension. From Figure 2, it can be determined that

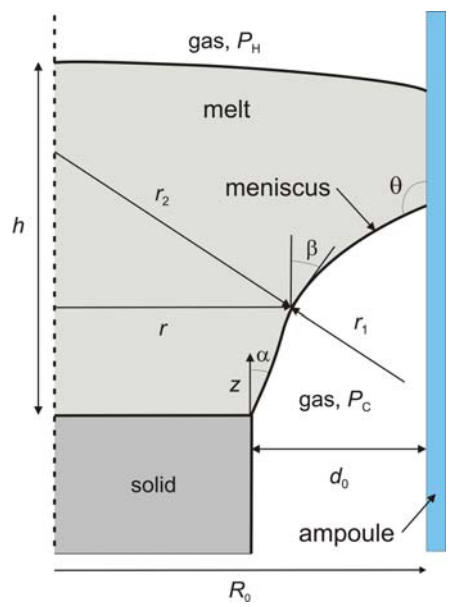

Figure 2. Detailed schematic of the meniscus region during detached solidification. Shown is the cross section of the cylindrical ampoule with the central axis of the ampoule on the left-hand-side of the figure and the ampoule wall at a distance R_0 on the right. *Note:* P_H, a pressure in the gas above the melt; P_C, a pressure in the gas below the melt; *h*, melt height; *z*, vertical coordinate; *r*, radial coordinate; α, growth angle; θ, wetting angle; d_0, gap distance from the wall; r_1, radius in the plane of the figure; r_2, radius in a perpendicular plane with its origin on the sample axis.

$$r_1 = \frac{r}{\cos\beta}, \tag{4}$$

where r is the radial distance measured from the centerline, and

$$r_2 = -\frac{1}{\sin\beta}\frac{d\beta}{dr}. \tag{5}$$

The boundary conditions on the meniscus are that $\beta = \alpha$ at the gap distance d_0 from the wall and that $\beta = \pi - \theta$ at the ampoule wall.

The solutions to the Young-Laplace equation can be divided into two cases: $\theta + \alpha < 180°$ and $\theta + \alpha > 180°$. Figure 3a shows the gap width versus pressure differential across the meniscus at the three-phase junction for the case where $\theta + \alpha < 180°$.[10] The material properties used to generate the curves are those for Ge and an inner ampoule diameter of 12 mm. The data shown are solutions for a stable gap width, in which the angle that the meniscus makes with respect to the vertical is also the growth angle α. For smaller angles, the gap would increase during growth, and for lower angles, the gap width would decrease, with the solid eventually running into the ampoule wall. A significant result is that this stable growth only occurs over the narrow pressure range of a few hundred pascals. For ΔP_0 greater than the range shown in Figure 3a, the gap width becomes comparable to the roughness of the ampoule, preventing complete detachment. The dashed line in Figure 3a demarcates the type of shape the menisci have, depending on the pressure ΔP_0. For $\Delta P_0 < -100$ Pa, the menisci have a convex/concave (with respect to the melt) interface shape, and for $\Delta P_0 > -100$ Pa, the menisci take on a purely concave shape. Representative menisci shapes for pressure differences of $\Delta P_0 = 0$ and -200 Pa are shown in Figure 3b.

The meniscus shape is important because it indicates under what conditions a stable gap width can exist. A Lyapunov stability analysis[6,7] found that the gap width will remain constant during growth only if $\partial\beta/\partial r < 0$ at the three-phase junction. That is, the meniscus must be convex at the solid-melt interface. This implies that stable growth can only exist for $\Delta P_0 < -100$ Pa where the gap widths are relatively large. This suggests why detached growth has been more readily observed in microgravity experiments. On Earth, gravity will cause the melt to run down into the gap if d_0 is larger than approximately 50–100 micrometers.

The gap width as a function of ΔP_0 for $\theta + \alpha > 180°$ is shown in Figure 4a. As compared to the case for $\theta + \alpha < 180°$, there is a broader range of pressure differential in which growth can occur with a stable gap width. The larger the wetting angle, the larger the gap width, and the more readily detachment can be achieved. For large negative values of ΔP_0 (several thousand pascal), the gap width becomes as small as the ampoule roughness, and detachment cannot be maintained. As ΔP_0 approaches zero, the gap width becomes large, which may result in melt rundown into the gap for the case of processing on Earth. The menisci shapes under these conditions are convex and are shown in Figure 4b. Because they are convex, they are stable according to the Lyapunov stability criterion. However, the degree of curvature is rather small, and so the driving force to return to the equilibrium gap width after a perturbation is also small.

Experimental Considerations

Crucible Material Choice

The first important experimental parameter is the choice of the crucible material. The wetting angle of the melt with the crucible material should be as high as possible. Because the wetting angle is a function of the interfacial energies solid-liquid (σ_{sl}), solid-gas (σ_{sg}), and liquid-gas (σ_{lg}) as given by the Young equation

$$\cos\theta = \frac{\sigma_{sg} - \sigma_{sl}}{\sigma_{lg}}, \tag{6}$$

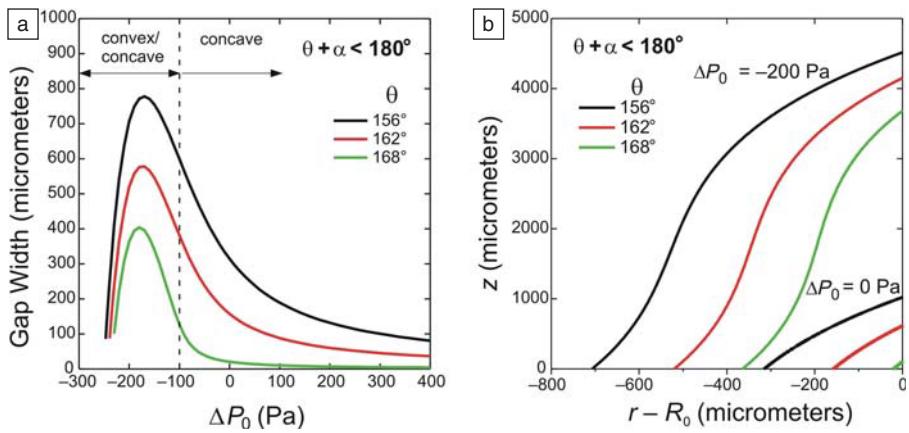

Figure 3. (a) Gap width as a function of meniscus pressure differential for $\theta + \alpha < 180°$. (b) Corresponding menisci shapes for $\Delta P_0 = -200$ and 0 Pa. The material properties used to generate the curves are those for Ge ($\alpha = 11°$) and an inner ampoule diameter of 12 mm. Data for three values of θ are shown.

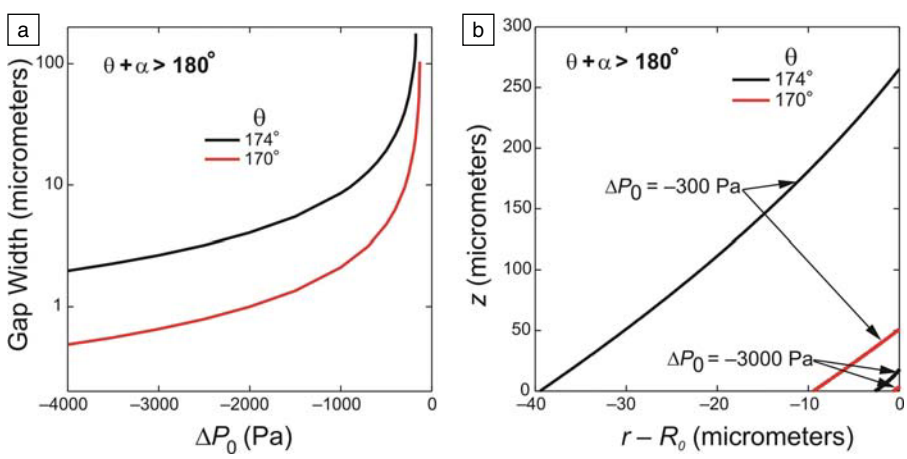

Figure 4. (a) Gap width as a function of meniscus pressure differential for $\theta + \alpha > 180°$. (b) Corresponding menisci shapes. The material properties used to generate the curves are those for Ge ($\alpha = 11°$) and an inner ampoule diameter of 12 mm. Data for two values of θ are shown.

materials with a low surface energy such as pyrolytic boron nitride (pBN) are preferable. However, any chemical reactions need to be precluded. Table I shows the wetting angles for germanium on several crucible materials as an example, with pBN and graphite having the highest wetting angles. However, the addition of even small amounts of silicon leads to a drastic drop of the wetting angle on the carbon-based substrate due to the formation of SiC, from 166° down to 70°.[12] Many times, reactions are much slower, but because crystal growth times can be of the order of days to weeks, even those slow reactions can change the wetting angle, as seen in the values for Ge on SiO_2 in Table I. Two other effects have to be kept in mind:

1. Because surface conditions control wetting, contamination even in the ppb range can have considerable effects. As an example, the wetting angle of Si on Si_3N_4 goes from 134° to complete wetting with a reduction of the oxygen partial pressure.[13] For Ge and Ge-Si mixtures, higher wetting angles are achieved through the use of oxygen getters or hydrogen.[12,14] In the case of GaSb in fused quartz crucibles, however, an increase of the wetting angle with a slight oxygen contamination has been reported.[15] This is probably due to a favorable effect of an oxide layer on the melt in this specific case.
2. The surface structure changes the apparent wetting angle; a rougher surface usually leads to an increase of the wetting angle if the original wetting angle on a smooth surface is greater than 90°. The disadvantage is, of course, that in the case of contact, a rough surface enhances the formation of secondary nucleation. It has been proposed to take advantage of the roughness effect by either structuring the ampoule[5] or, in the case of CdTe, using carbon-based "velvet" as crucible cladding.[16]

Table I: Wetting Angles of Germanium on Different Substrates.

	Substrate	Surface Treatment	Atmosphere	Wetting Angle Ge (degrees)
O-Based Substrates	Fused silica		Vacuum	150–117*
		Sandblasted	Vacuum	155–128*
		Fume etched	Vacuum	156–131*
		Fire polished	Vacuum	148–131*
	Sapphire		Vacuum/argon	150–119*
		Sandblasted	Vacuum/argon	144–134*
C-Based Substrates	Graphite		Vacuum	166
	Glassy carbon		Vacuum	157
	CVD diamond		Vacuum	146
	SiC		Vacuum	166
	C-based aerogel		Vacuum/argon	171
N-Based Substrates	Si_3N_4		Vacuum/argon	157–136*
	AlN		Vacuum	170–153*
	pBN		Forming gas	172

* Denotes a change of the angle over time due to a reaction of the substrate.
Note: CVD is chemical vapor deposition.

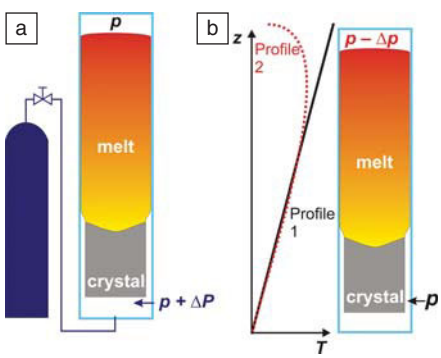

Figure 5. Two possibilities to achieve an overpressure in the gap, which can promote detachment: (a) An external gas source and (b) a change of the temperature profile after melting but before growth.

Generation of a Pressure Difference

Once the crucible material is chosen, it is advantageous during processing on Earth to allow some overpressure in the gap, as shown previously in the theory section. The overpressure cannot be too large, as otherwise, and depending on the gap width, bubbling will occur and limit the pressure difference. There are basically three ways to achieve a pressure difference (two of which are shown in Figure 5):

1. The use of an external source of pressurized gas (Figure 5a). Duffar et al.[7,17] have used and described this approach. It works, but the two main drawbacks are the nearly inevitable contamination of the system, even with highly purified gases, and the difficulty of controlling the pressure finely enough to keep the meniscus in the stable regime.[17]
2. Changing the axial temperature gradient in a multizone furnace after melting but before growth,[17–22] as shown in Figure 5b. The crucible is filled with an inert gas and must be closed at the bottom. The starting material is molten using a normal Bridgman temperature profile (Profile 1). The melt now seals the volume around and below the seed area from the gas volume above the melt. At this point, the pressure above and below the melt is still nearly equal. Now the temperature is reduced in the gas volume above the melt (Profile 2), leading to a pressure reduction in that area and thus a relative overpressure in the volume below the melt. With this method, it was possible to achieve detached growth on Earth for Ge, Ge_xSi_{1-x},[18–22] and GaSb.[17] It is interesting to note that the detrimental influence of gravity on detachment has been demonstrated using the Ge-pBN system. Ge can be grown detached in pBN reproducibly by using the temperature profile inversion method; even just growing in a closed-bottom crucible is often enough to achieve detachment on earth. The same experiment in an open-bottom crucible, where no pressure difference is possible, leads to attachment in the melt. Growing in an open-bottom crucible under microgravity, however, leads to full detachment (Figure 6).
3. The use of an internal pressure source below the melt. This is feasible for compound materials where one component has a considerably higher vapor pressure over the element than over the compound, such as As in the case of GaAs or Cd in the case of CdTe. This approach influences the stoichiometry somewhat, but this actually can be positive (e.g., to

Figure 6. Ge-Si crystals (diameter 15 mm) grown from a $Ge_{0.98}Si_{0.02}$ melt in an open pBN crucible (a) under microgravity during the Foton M3 mission and (b) on the ground. The 1-*g* (one-gravity) sample is completely attached; the microgravity one fully detached over 20 mm.

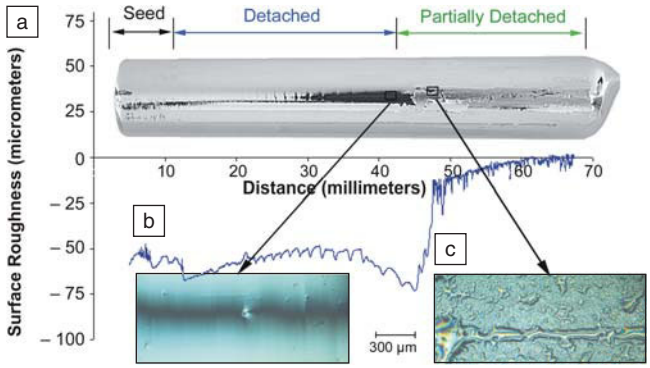

Figure 7. (a) Ge crystal grown by the Bridgman method in a pBN crucible, grown detached for the first 40 mm and mostly attached for the last part (from the experiment series described in References 18–21 the growth direction is from left to right). The graph below the crystal shows a profilometer measurement of the surface, indicating a gap width of about 60 µm in the detached part. The micrographs at the bottom show the different surface appearance of the detached and attached parts: the detached part (b) shows a shiny surface with growth lines similar to CZ or FZ crystals, whereas the attached part (c) shows a rough surface that is a negative image of the structure of the pBN crucible.

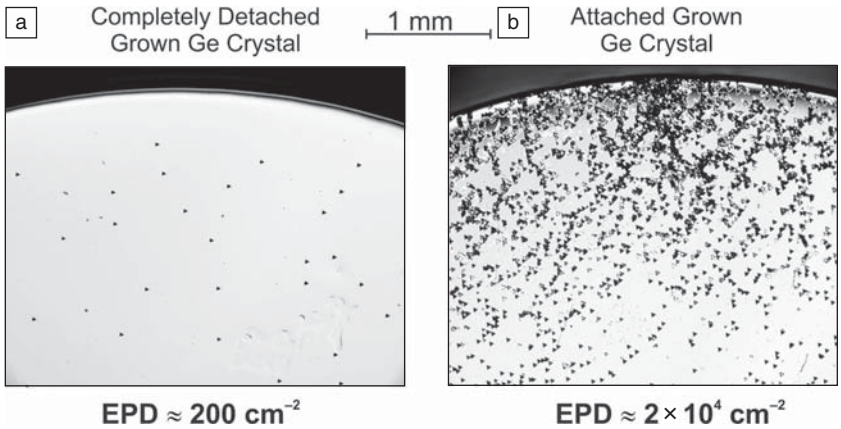

Figure 8. Etch pits in wafers cut from (a) a detached grown Ge crystal grown in microgravity (µg) and (b) an attached Ge crystal grown at 1g. The etch pit density in the outer parts of the detached grown crystal is about two orders of magnitude lower than in the attached grown one.[19] *Note:* EPD is etch pit density.

reduce the occurrence of tellurium inclusions in CdTe). This method has been successfully used for CdTe.[23]

Detached Growth Results

Figure 7 shows a typical example of a germanium crystal that was grown detached in the first part and mostly attached in the second part, together with profilometer measurements and surface micrographs. The profilometer measurements show a gap width of approximately 60 µm. The detached grown part looks like a CZ or a FZ crystal with a shiny surface and growth lines, which is quite characteristic for detached grown Ge. The attached part shows the rough surface structure of the pBN crucible. Note that the difference in appearance depends on the materials; for other combinations, such as CdTe in a fused quartz ampoule, the detached part has a rougher appearance than the attached part. The advantages of detached growth are shown in Figures 8 and 9, comparing attached and detached grown Ge crystals.[19] The only difference between the growth procedures was the use of an open-bottom crucible in the case of attached growth. The etch pit density is up to two orders of magnitude smaller in the detached-grown crystal. These results have later been corroborated by synchrotron topographs showing a reduction of dislocation densities from 10^4 cm^{-2} in the seed to 10^2–10^3 cm^{-2} in the detached crystal, whereas for the attached crystal, the dislocation density stayed at 10^4 cm^{-2} or increased to 10^5 cm^{-2} at the periphery.[24] In addition to the direct improvement of the structural quality by detachment, there is also a secondary effect that can be positive. Because of the gap between the crystal and the wall, the heat transfer to the wall is reduced, resulting in less interface curvature and lower stress in the crystal. As an example, for a 2-in. diameter VGF-grown germanium crystal, a reduction of the interface curvature from 13.4% to 10.1% and a resulting reduction of the von Mises stress by 30% has been calculated.[22]

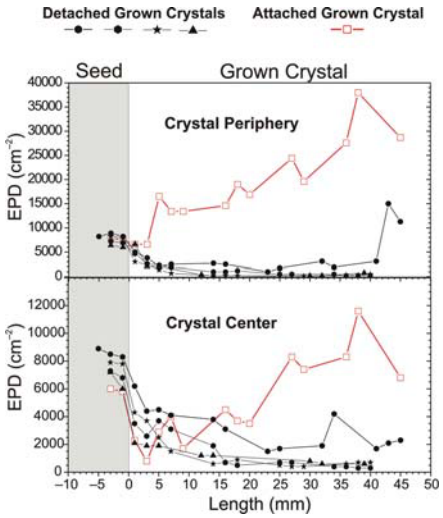

Figure 9. Etch pit density versus crystal length for several detached grown Ge crystals (black curves) and an attached grown crystal (red curve). The graphs show (a) the values for the periphery of the crystals, and (b) the center values.[19] *Note:* EPD is etch pit density.

Summary

The detached or dewetted growth of crystals has come quite a way from a strange effect in space experiments more than 30 years ago to a possible tool in the arsenal of the crystal grower that has been shown to improve crystal quality considerably—on Earth! In the last decade, considerable progress has been made in the theoretical understanding of detachment as well as in experimental procedures, although much more work needs to be done. Based on the present state of the art, detachment can be achieved reproducibly for systems where the sum of the growth and wetting angle is close to 180° and with the help of a slight overpressure in the gap below the melt, preferably through control of the temperature field. The major obstacles arise from the fact that detachment can be very sensitive to contamination at the ppb level, and for stable detachment, the meniscus shape (i.e., the pressure difference) needs to be finely controlled.

Acknowledgments

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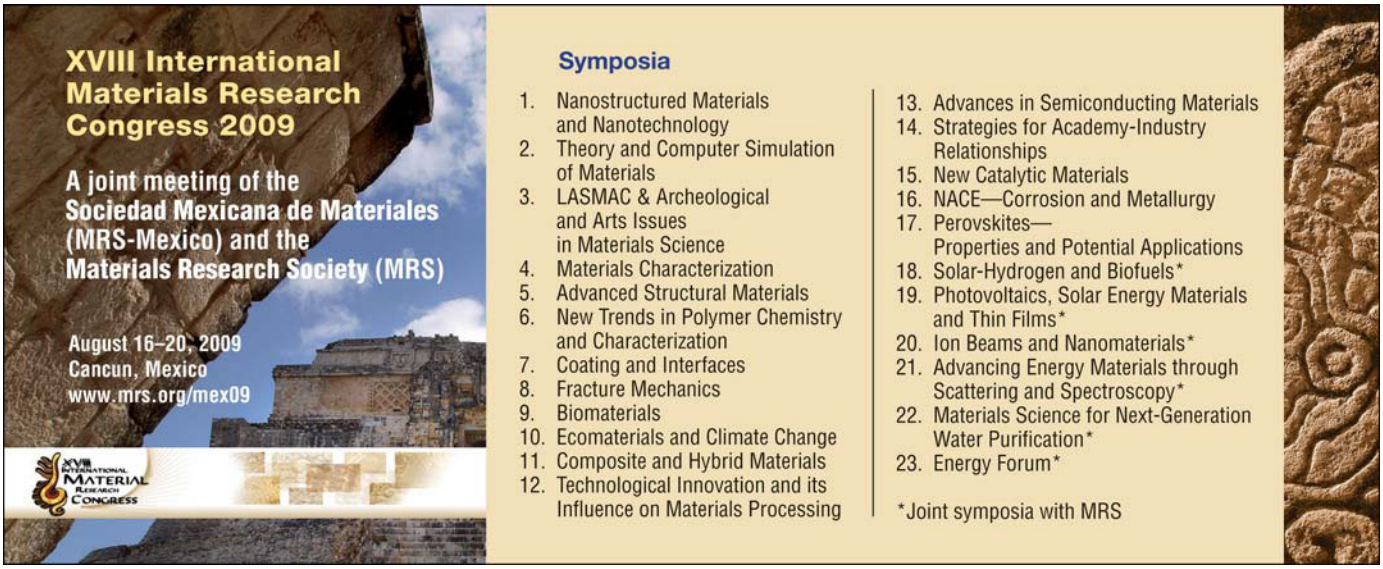
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